Curriculum Vitae

Personal data:

Names and Surname: Paweł Karol Jastrzębski
Location: Warsaw, Poland
Telephone: E-mail: (+48) 502–057–486
Date of birth: 21 January 1987

Education:

2007–2014	Warsaw University of Technology, Faculty of Electronics and Information Technology, Computer Science, BSc. and
MSc. 2010–2011	Universidad Politécnica de Madrid, Facultad de Informática, Ingenería Informática
2003–2006	XLI High School of Joachim Lelewel in Warsaw, specialization of class: mathematics and physics

Preferred technologies:

C/C++, embedded systems, Linux, Vulkan, Git, game programming

Work experience:

03.2024–currently — Samsung R&D Institute Poland

Senior Software Engineer:

- Development and architecture design of OpenVR/OpenXR solutions for DX11, DX12 and Vulkan for a R&D project: new features, no legacy code. Performance analysis for UEVR solutions. Working in international Polish-Korean team.

Tools and technologies used: **C/C++17**, **CMake, DX11, DX12, Vulkan**, **github**, **MSVC**, **Jira**, **Git**, **UEVR**, **RenderDoc**, **nVidia Nsight, PIX**

09.2020–03.2024 — Mobica Ltd. Sp. z o. o.

Senior Software Engineer:

- Development and integration of software stack for virtualization Vulkan API for embedded automotive solutions.
- Development and integration of GPU driver and performance measurement tools for smartphone solutions.
- Development and integration of graphics stack for IVI automotive system based on OpenGL and Vulkan.
- Development of Open-source project of OpenCL-CTS test suite.
- Development of Open-source project of Vulkan Samples suite.

Tools and technologies used: **C/C++**, **Linux**, **Vulkan**, **shell scripting**, **code review** (gerrit)/(gitlab), **Jira**, **BitBake**, **gerrit**, **Git**, **CI** (Jenkins), **RenderDoc**, **nVidia Nsight**

05.2020–09.2020 — Huuuge Games

C++ Developer:

- Creating frontend and backend of multi-platform social mobile game, creating a multi-platform build system.

Tools and technologies used: **C/C++**, **Lua**, **Android**, **Docker**, **code review** (Git), **Jira**, **Git**, **CI** (Jenkins), **Grafana**

07.2019–05.2020 — intive GmbH

Software Engineer:

- Integrating software for DVB digital TV STB solutions (x86). Fixing bugs, Linux drivers maintenance, kernel features integration, system metrics analysis, code review, static code coverage. Business trips to the client's sites in Poland and abroad.

Tools and technologies used: **C/C++**, **Linux**, **shell scripting**, **RDK**, **code review** (gerrit), **Jira**, **BitBake**, **gerrit**, **Git**, **CI** (Jenkins), **Grafana**

06.2018–06.2019 — Cubiware / TiVo Poland Sp. z o.o.

Senior Software Engineer:

- Developing software for DVB digital TV STB solutions (MIPS, ARM). Legacy code maintenance, developing code for new products, MPEG TS parsing, unit tests, component tests, code review, static code coverage, cross compilation for various platforms.

Tools and technologies used: **C** (with own Object-Oriented extensions), **Linux**, **gcc**, **clang**, **gdb**, **Valgrind, Visual Studio Code**, **UML**, **SCRUM**, **Mercurial**, **code review** (Code Collaborator), **Jira**, **Confluence**, **Git**, **CI** (Jenkins)

07.2016–03.2018 — QLOC S.A.

Porting Developer:

- Porting AAA computer games for various platforms (PC/consoles/mobile devices)
- Games ported:
  - WWE2k17 (PS4 to PC)
  - Dark Souls: REMASTERED (PS3 to PS4/Xbox One/PC)

Tools and technologies used: **C/C++**, **Windows**, **Visual Studio 2015**, **Perforce**, **code review** (Code Collaborator), **Jira**, **CI** (Jenkins)

08.2014–07.2016	Intel Technology Poland	Graphics Software Engineer: • Development of graphics cards drivers. Area of responsibilities: maintaining the code of existing **OpenGL ES** driver for **Android**, **Windows** and **Linux,** adding new features to the existing driver, fixing bugs. • Leading an international team developing ASTC and hardware tessellation demos for **SIGGRAPH'2015**. • Porting existing graphics driver to **Integrity RTOS**. Tools and technologies used: **C/C++**, **Windows**, **Linux**, **gcc**, **gdb**, **Valgrind, Visual Studio 2012, Perforce**, **code review** (Code Collaborator), **CI**, various benchmarks and test applications.
10.2013–07.2014	Warsaw University of Technology	Software Engineer: • Image correlation science project. Developing new improvements and optimizing the time and memory complexity of the rendering part of the project. Implementing tests to check the new features. Tools and technologies used: **C++**, **Qt5**, **Windows**, **Visual Studio 2012**, **Mercurial**
03.2011–09.2011	Accenture	Technical Application Team Member: • Administration of **Linux** and **Unix** based systems. Mangment of a big telecommunication system. Developing unit tests. Tools and technologies used: **Linux**, **Unix**, **WebSphere**, **JBoss**, **Python**, **Oracle DBMS**, **PostgreSQL**, **Trac**, **SVN**

Skills:

Programming (C/C++, OpenGL (ES), GLSL, Android)

Scrum/Agile, Continuous integration

Version control systems (Git, Mercurial, Perforce)

Information/task maintenance (Jira, Confluence)

Operating systems (MS Windows, GNU/Linux, Android)

Conferences:

On game start'2013	Participated in an international conference about developing computer games (http://ongamestart.com/).
SIGGRAPH'2015	Presented several OpenGL ES 3.2 demos on Intel CHT tablet (https://www.youtube.com/watch?v=gCYpyNjDmro).
Vulkanised'2023	Participated in an international conference for development using Vulkan API (https://www.vulkan.org/events/vulkanised-2023)

Competitions and prizes:

One week compo'2003	1st place
IGK'2015	2nd prize
Global Game Jam	-

Foreign languages:

English	fluent in speech and writing (FCE and CAE certificates)
Spanish	communicative in speech and writing
Polish	native speaker

Additional skills:

Driver's license, category B

Yacht sailor's license

Interests:

Computer Science

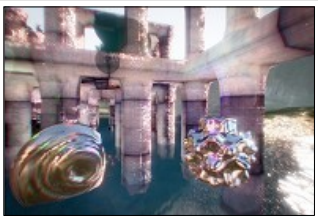
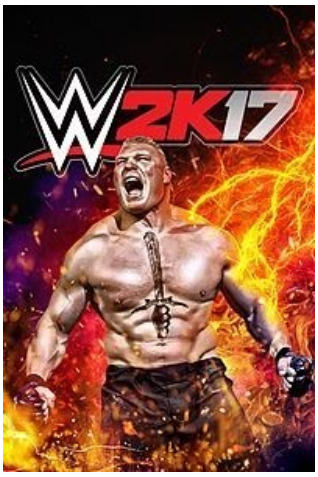
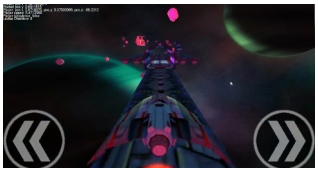
Physics and astronomy

Spanish and South American culture

Economy and trading

Wyrażam zgodę na przetwarzanie moich danych osobowych dla potrzeb niezbędnych do realizacji procesu rekrutacji (zgodnie z Ustawą z dnia 29.08.1997 roku o Ochronie Danych Osobowych; tekst jednolity: Dz. U. 2016 r. poz. 922).

Professional and hobby projects:

	07.2018–2021	Vulkan Based 3D Rendering Engine	C++, Vulkan, GLSL, Qt	Hobby project developed in spare time as self-improvement activity. The main aim of this project is to create a rendering engine for real-time procedural 3D content generation using the newest possible technologies.
	07.2017–03.2018	Dark Souls: Remastered	C++, DirectX, PS4, Xbox One, PC	Member of a porting team of Dark Souls: Prepare to Die to PC/PS4/Xbox One. My tasks were very diversified: sound system, input, gameplay, physics, animations and some performance optimizations.
	08.2016–05.2017	WWE2k17	C++, DirectX	Member of a porting team of WWE2k17 from PS4 to PC. My main tasks were adjusting menus for the PC version of the game. I was also working on some speed optimizations and gameplay bugs.
	05–08.2015	Demos for SIGGRAPH'2015	C++, OpenGL ES, Android	Coordinating a Polish – Israeli team working on OpenGL ES 3.2 demos. Our demos were presented ASTC compression and hardware tessellation. Both demos were shown by myself at SIGGRAPH'2015 in Los Angeles, CA.
	05.2013–06.2014	3D Arcade game	Andoid, Java, OpenGL ES, libGDX	A commercial hobby project. I was involved as a manager and programmer of this project. The team consisted of 4 members: 2 programmers and 2 graphic designers.
	05–06.2012	Jumping Kitties Visual	C++, MS Studio, Marmelade	A commercial arcade/platform game for most common mobile platforms (iOS, Android, Windows Phone). The game was developed with C++ using Marmelade.